OCSA · PRODUCTION & DESIGN CONSERVATORY

ALEXANDER KIM

PORTFOLIO

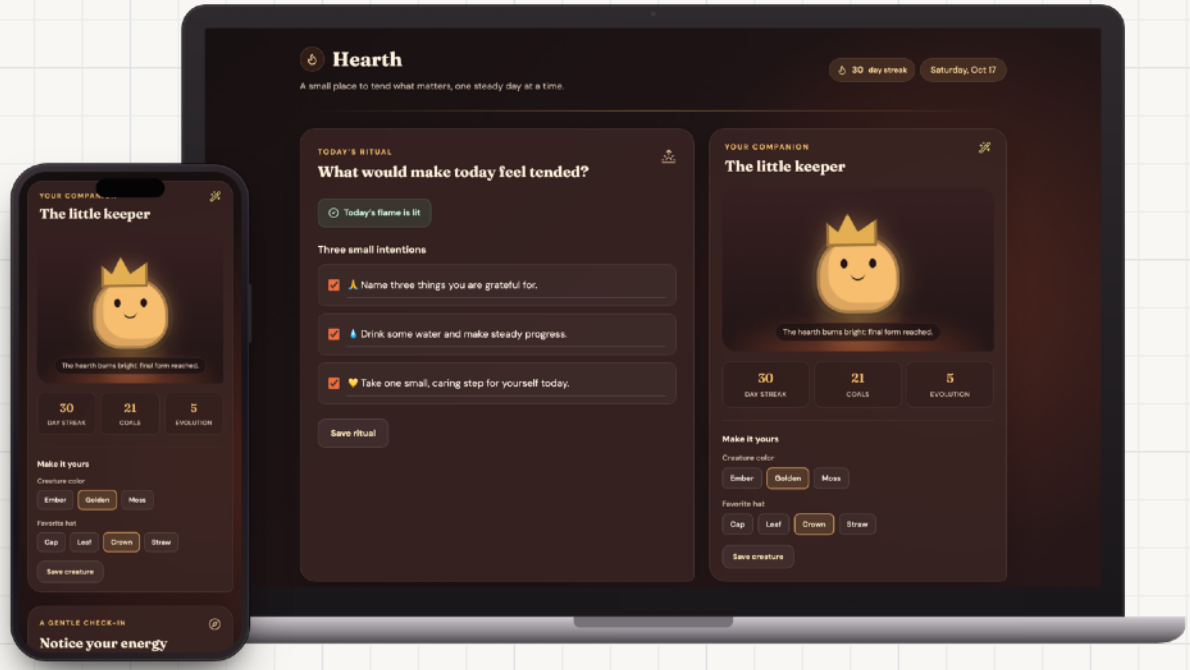
stuff I build, code, draw and design

02 Hearth (app)
05 Go-Kart Build
06 Meet the Crew
07 Go-Kart World
08 LEGO Virtual Portfolio
09 Wheelie Ride
10 The Lobby
11 Mercedes SL63
12 Poppin' Matcha Booth
13 Poppin' Matcha Ads
14 Alex & Mr. Fishy
16 The Fort
17 Birdhouse & Glider
18 Music Production
19 Encanto



HEARTH

a wellness app that grows with you



▪ WHAT IT IS

Hearth is a wellness app built around a daily ritual: each morning it asks “What would make today feel tended?” and you set three small intentions. A quick check-in on your mood, energy and stress shapes kinder goals, an “I’m feeling” page suggests ways to cope, and you get a companion that grows with your streak.

▪ MY ROLE

I was the designer and coder: I came up with the idea, designed every screen and feature, and coded it by directing an Agent coding tool for the OC Student AI Network.

▪ WHY I INCLUDED IT

Like a lot of students, I juggle assignments, deadlines and social pressure, and I wanted a calm way to take each day one step at a time. I noticed a lot of people make games, so I wanted to make something that stood out and could actually help.

3
INTENTIONS A DAY

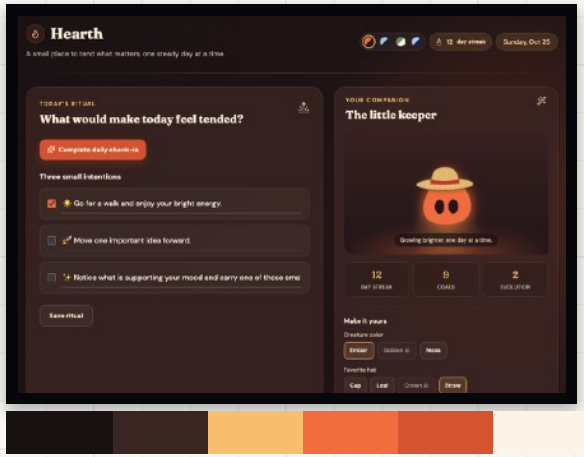
28
DAYS TO THE FINAL FORM

no ads, no tracking!

0
DATA COLLECTED

COLORS THAT MATCH YOUR MOOD

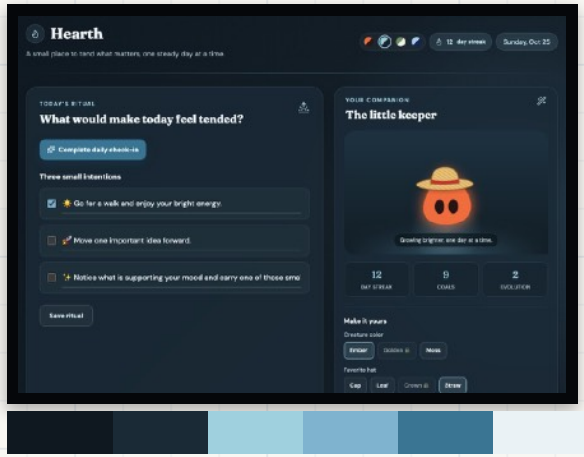
Colors change how people feel, so I designed four calm themes you can switch between depending on your mood. Each one comes from a place or memory that makes me feel calm.



EMBER

cozy + comforting

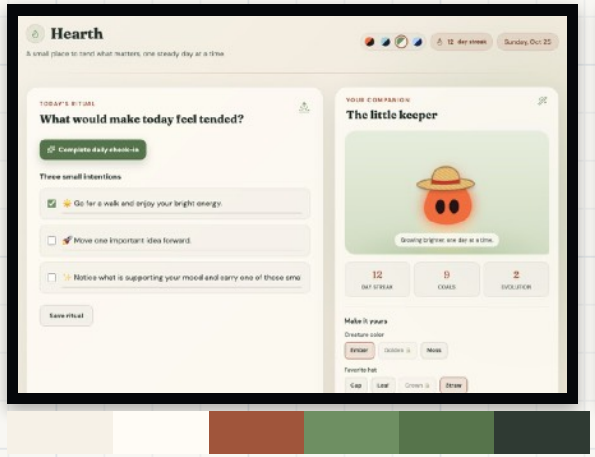
From camping trips with my mom, where the campfire is one of the most comforting feelings I know.



TIDEPOOL

serene + connected

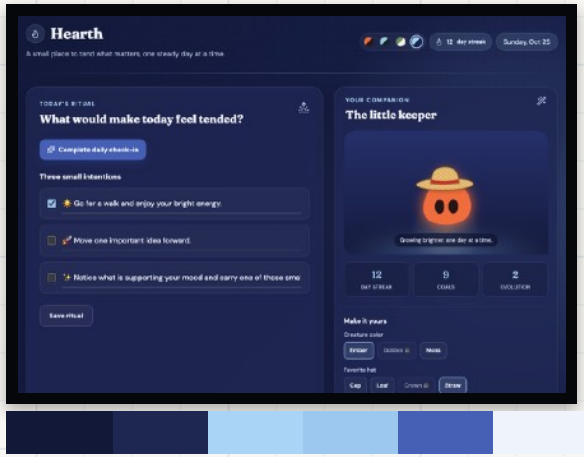
From exploring tide pools as a kid, where every animal was relaxed and the ecosystem worked as one.



MORNING MEADOW

fresh + hopeful

From mornings lying in the grass under a blue sky, with nowhere to be and a whole new day ahead.



MARSHY

soft + safe

From Marshy, my sequined midnight-blue sheep, who kept me company on the darkest nights.

TURNING PROBLEMS INTO SOLUTIONS

1 THINK

I sat down and asked myself what tool could make the world better and isn't out there enough.

2 OUTLINE

I wrote the main outline: a daily check-in, three goals, and a creature that grows.

3 BUILD

I built it by prompting an Agent coding tool, one feature at a time.

4 TEST & FIX

I tested each part, and when something broke I found another way.

WHAT DIDN'T WORK, AND WHAT I DID ABOUT IT

The app wouldn't load

My first version wouldn't run at all.

Fix: I loaded it on a personal account, and it worked.

FIXED

The Agent got stuck on goals

It kept misreading the goals feature and handing back the same broken answer, no matter how I prompted it.

Fix: I turned the problem into the solution: a goal tracker you set up yourself with short surveys.

FIXED

The shop locks free hats

The Agent decided you have to buy the free hats for your creature.

Fix: Still working on it, and it's next on my list.

IN PROGRESS

GO-KART BUILD

WHAT IT IS

This summer I went to a go-kart camp and built my own kart.

MY ROLE

I built the frame, added a stick shifter, and painted it. My favorite car is the Nissan Skyline, so I based the look on Brian's Skyline from Fast & Furious: silver with blue stripes, plus blue underglow lights to make it more realistic.

WHY I INCLUDED IT

It's the first real vehicle I've built, and it's what inspired my racing game.

THE SKYLINE LOOK

- Silver body
- Navy stripes
- Blue underglow



4



4 GTR = the car name

1



1 the frame

3



3 paint + stripes

2

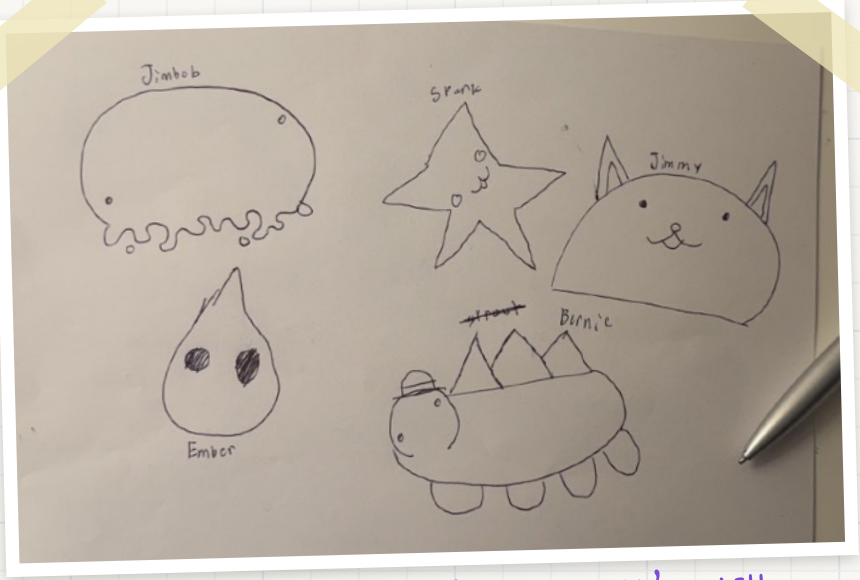


2 stick shifter!

MEET THE CREW

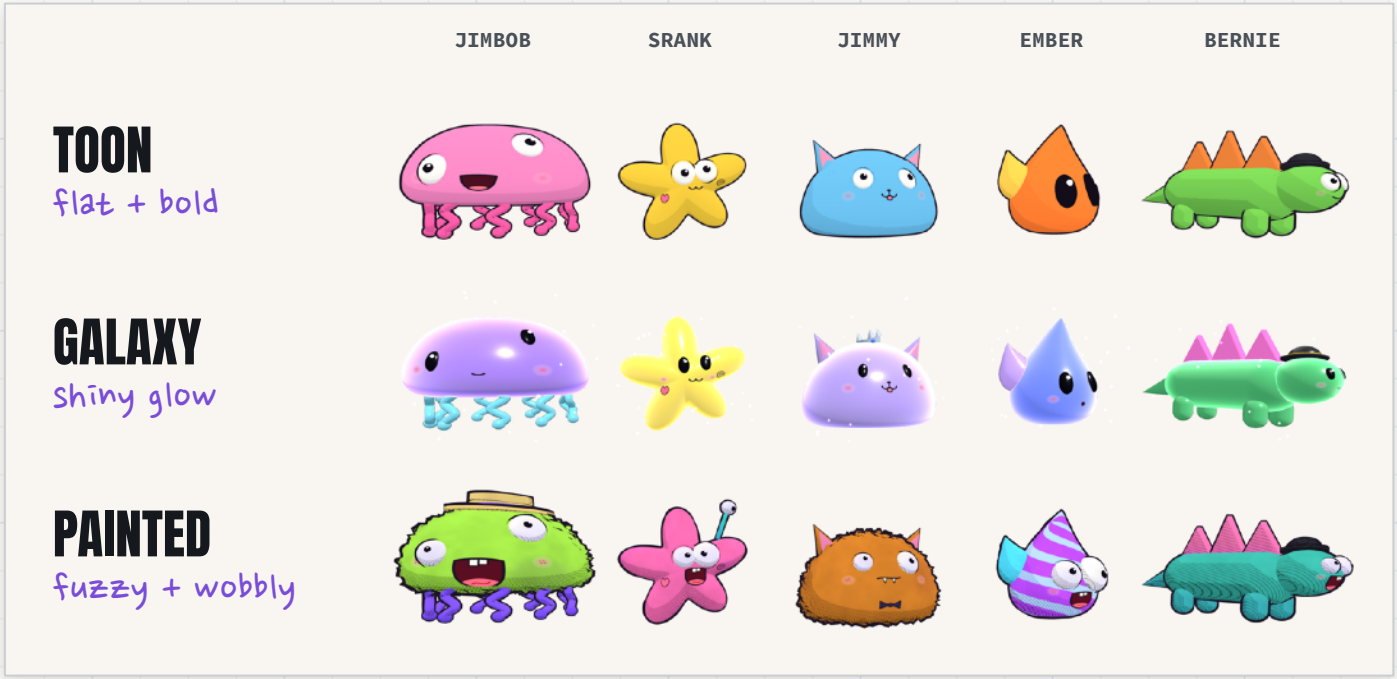
Jimbob, Srank, Jimmy, Ember and Bernie are five characters I invented and first drew in pen on paper. I turned each sketch into a 3D character, then tried every one in three looks inspired by cartoons and games I studied. I included them to show my process: how a quick doodle grows into a finished character without losing what made the drawing special. They're the racers in my go-kart game.

1 SKETCH ON PAPER



Bernie was almost named Sprout. It's still crossed out!

2 BRING THEM TO LIFE IN 3D, IN THREE LOOKS



07 CODING · GAME DESIGN

GO-KART WORLD

WHAT IT IS

I'm really into racing games, so I made a 3D racing game inspired by the go-kart I built and the Crazy Cart I'm customizing.

MY ROLE

I drew the characters by hand, then built the game in Visual Studio Code using an agent orchestration harness with the Opus 5.5 model. I added different modes, a drift track and details like sponsor banners inspired by real racing brands, plus Brian's Skyline turning on a stand in the middle.

WHY I INCLUDED IT

It shows how I turn something I built in real life into something people can play.

MODES + FEATURES

RACE THE CREW | FREE DRIVE | DRIFT POINTS | NIGHT | 4 CAMERAS | VR | RACE HUD







drifting past the crew

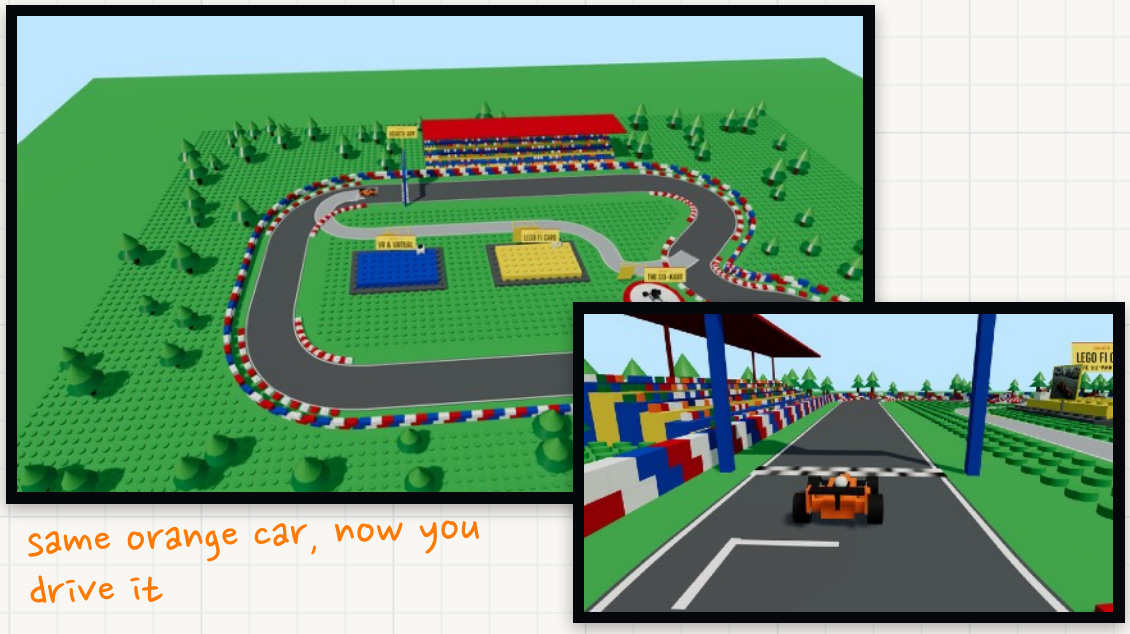
LEGO VIRTUAL PORTFOLIO

▪ REAL BRICKS



I shot them at street level so they look real

▪ VIRTUAL CIRCUIT (BRICK BUILT)



same orange car, now you drive it

▪ WHAT IT IS

I built two LEGO F1 cars and photographed them outside at street level, so they look like real race cars.

▪ MY ROLE

Then I used them to design a virtual portfolio: a LEGO-scale 3D race track where you drive my orange car and pull in to see each project.

▪ WHY I INCLUDED IT

It shows how something I built with my hands became the idea for a bigger digital world.

09 CODING · GAME + REAL MAP DATA

WHEELIE RIDE

WHAT IT IS

A 3D game where you pick a bike and hold a wheelie as long as you can without looping out. Version 1 was one straight road, so for Version 2 I rebuilt it in my own Orange County neighborhood with real map data.

MY ROLE

I designed and built both versions, and I wrote down every problem and fix in a release log.

WHY I INCLUDED IT

It shows how I keep improving my projects after version 1.

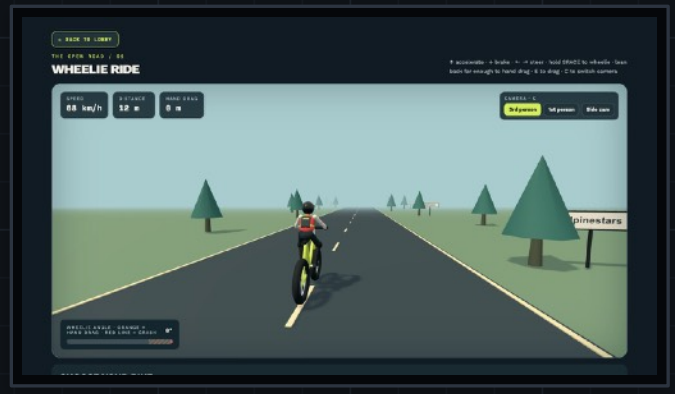
WHAT I LEARNED (FROM MY RELEASE LOG)

1 **Test your data, not just your code.** The ocean came out as dry land because the elevation data said the sea was 0 m high, same as a beach.

2 **Faking movement is easy until you want to go somewhere.** A real position and heading made maps, traffic and landmarks possible.

3 **Always test the extremes.** I only found a camera bug by testing the fastest bike.

VERSION 1 · ONE STRAIGHT ROAD



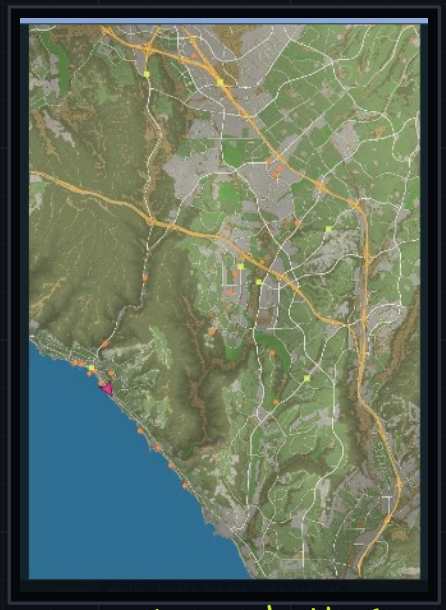
15,900 REAL ROADS

104,600 REAL BUILDINGS

15 × 21 km OF MY REAL MAP

VERSION 2 · MY NEIGHBORHOOD IN ORANGE COUNTY





click a road to ride there

REAL GEARS | JUMPS | CRASH PHYSICS | LIVE SOUND

10 CODING · 3D / VR CONCEPT

THE LOBBY



WHAT IT IS

A 3D game room for all my projects. Each project is a game cartridge you plug into my ALEX-2600 console, then you step through a portal into the game. Bit, a little robot, shows you around.

MY ROLE

I sketched this concept to showcase my work, then turned it into a 3D experience you can explore in a browser or a VR headset.

WHY I INCLUDED IT

I love the idea of a gaming system that plays the games I'm building, with a virtual robot as your guide.

HOW IT WORKS



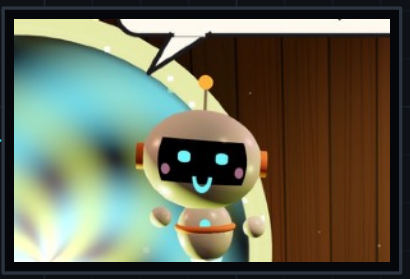
1 PICK A CARTRIDGE

Every project is a cartridge on the rack. The "?" slots are for games I haven't made yet.



2 PLUG IT IN

The ALEX-2600 console plays the game's title screen on the TV.



3 STEP THROUGH

Bit tells you about the game, then a portal opens straight into it.

11 CAD • 3D MODELING

MERCEDES SL63

WHAT IT IS

This is my 3D model of the Mercedes-AMG SL63, a famous roadster, made in Tinkercad.

MY ROLE

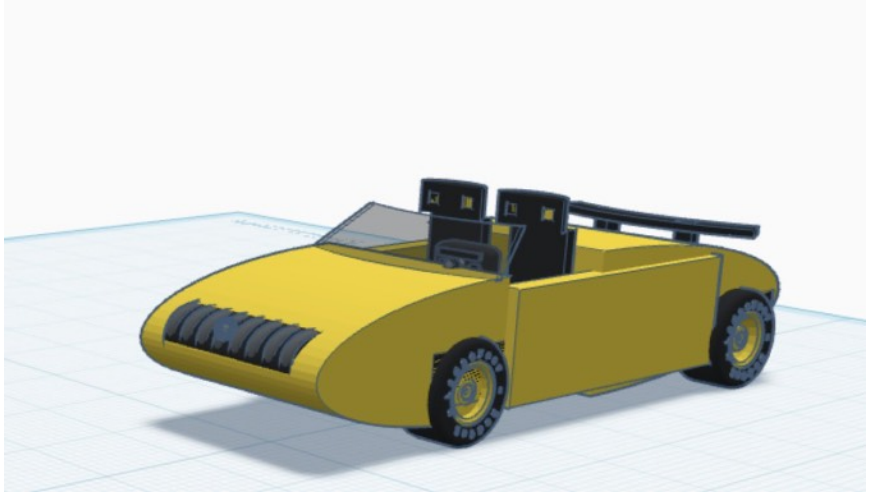
I'm a big fan of cars and I like designing in Tinkercad, so I made this as practice and as a 3D model for a game I'm making. I built it from simple shapes, down to the seats, the grille, the wheels and a steering wheel full of buttons.

WHY I INCLUDED IT

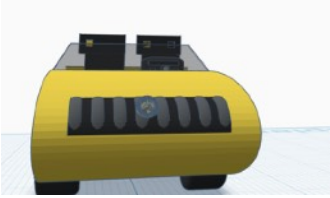
It shows my sense of scale and design, how I assemble a model from parts, and how I'm learning spatial awareness.

MODEL	TOOL
Mercedes-AMG SL63	**Tinkercad**
DRAWN BY	FOR
Alexander Kim	**my racing game**

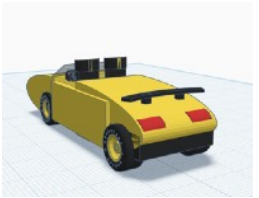
VIEW 1 • THREE-QUARTER



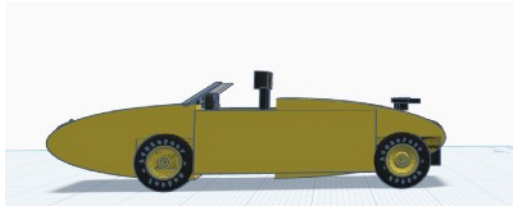
VIEW 2 • FRONT



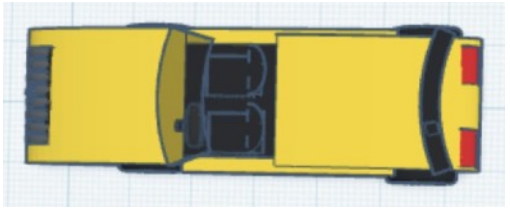
VIEW 3 • REAR



VIEW 4 • SIDE

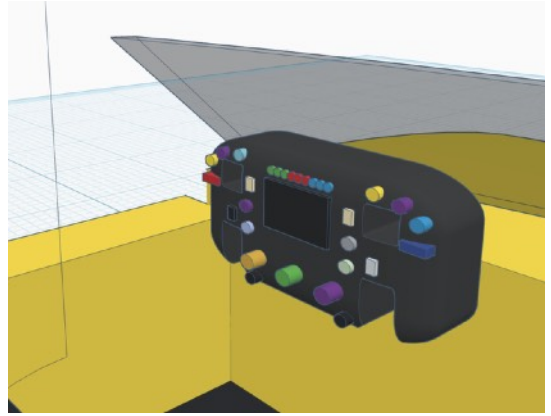


VIEW 5 • TOP



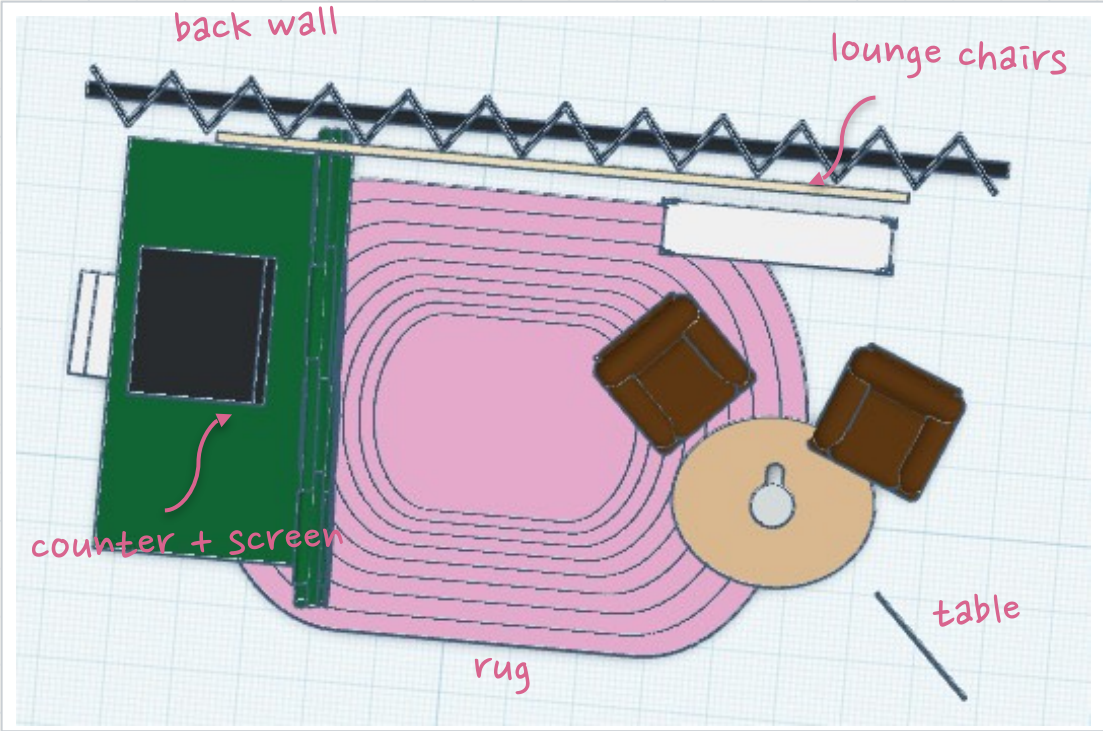
DETAIL A • STEERING WHEEL

even the wheel has buttons!



POPPIN' MATCHA BOOTH

PLAN • FROM ABOVE



FRONT



SIDE



WHAT IT IS

This is a booth I designed in CAD for Poppin' Matcha, the company my 7th and 8th grade class ran for Virtual Enterprises Junior Ventures.

MY ROLE

I was a Senior Manager in Marketing, responsible for ads and promotions.

WHY I INCLUDED IT

It shows how I design real-world environments in CAD, from the back wall and shelves to the lounge chairs and rug.

MY TEAM ROLE

SENIOR MANAGER, MARKETING

POPPIN' MATCHA ADS

WHAT IT IS

These are ads and promotions for Poppin' Matcha: a web banner, a vertical video ad and slides from our marketing plan.

MY ROLE

My role was helping create ideas for them. I used Canva, starting with templates and then customizing them for our brand.

WHY I INCLUDED IT

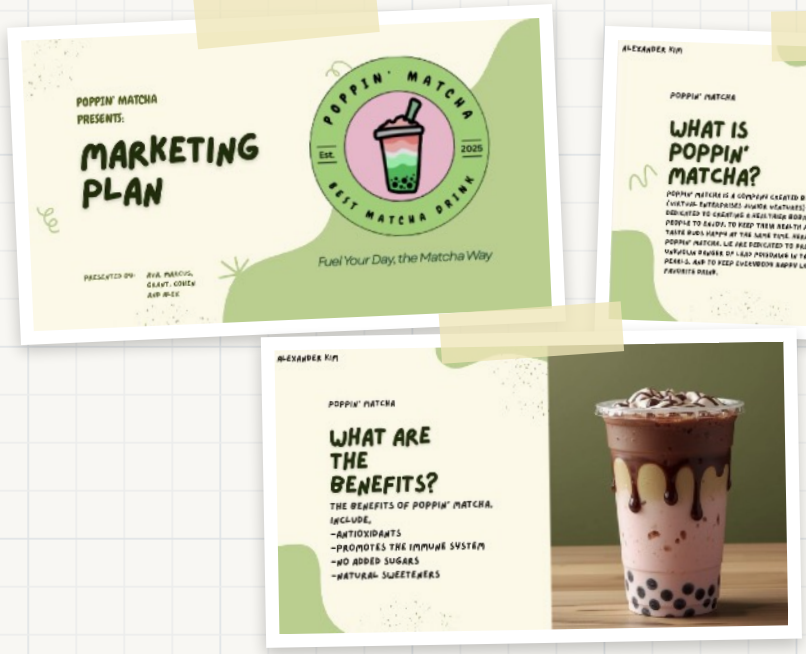
It shows graphic design for a brand our class built together as a team.

OUR BRAND MISSION

a boba with healthy benefits that still keeps its flavor

web banner





slides from our marketing plan



vertical video ad

ALEX & MR. FISHY

Alex & Mr. Fishy is my cartoon series. In every episode, Mr. Fishy gets into a mess and learns something with help from his friend Alex. I picked frames from each cartoon and laid them out as storyboards.

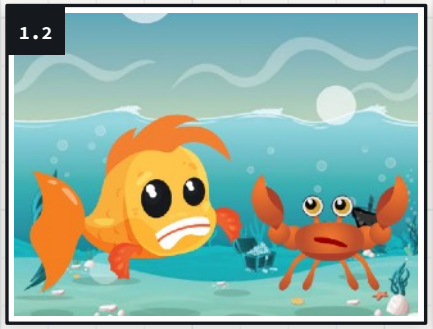
▪ EPISODE 1

FREAK OUT

A storm is coming and Mr. Fishy is scared. His friend Alex comes along and helps him discover he can trust that everything will be OK, and he can ride the waves.



Title card: a giant wave is coming!



Mr. Fishy freaks out about the storm.

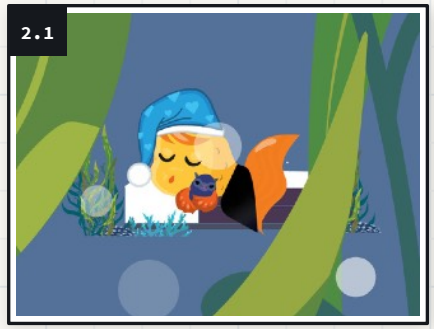


With Alex's help, he rides the waves.

▪ EPISODE 2

WAKE UP

Mr. Fishy and Alex help a school of fish rebuild their home after they made a mess "messing around." Brick by brick, they build it back up together.



Mr. Fishy is fast asleep.



His friends need him to wake up.



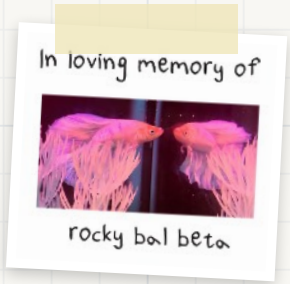
The fish's home is a mess of bricks.



Brick by brick, they build it back up together.

THE BIG RACE

Rocky Bal Beta was our real pet betta fish



▪ EPISODE 3

RACE

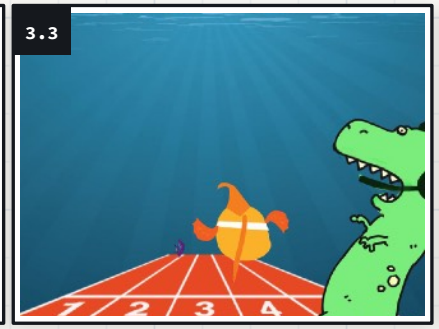
Mr. Fishy is sure he's the best and can win his race without training. He goes up against Rocky Bal Beta, who was our pet betta fish in real life. He loses, and learns that he needs to put the work in and practice humility.



Mr. Fishy, the "lean mean swimming machine."



His rival: Rocky Bal Beta.



Race day: he falls way behind.

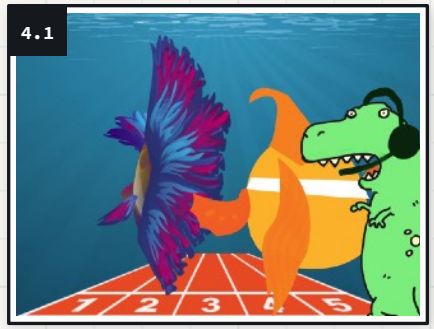


Time to put the work in.

▪ EPISODE 4

STRONGER

Mr. Fishy worked hard with Alex to get ready for the race, and he gets a rematch, with a lot of chaos.



The rematch!



Breaking news: "Banana Pants" interrupts the Tour de Fish.



A virus makes the computer sick.



And the chaos keeps coming.

THE FORT

WHAT IT IS

At a camp I went to with my brother, we built a fort out of wood.

MY ROLE

We used posts for the frame, built the first floor above the ground using nails, built the walls, added a roof, and then decorated it. My role was assembling the fort.

WHY I INCLUDED IT

It's where I got my experience building with wood.

HOW WE BUILT IT

1. **POSTS** — for the frame
2. **FLOOR** — nailed together, above the ground
3. **WALLS** — built and nailed on
4. **ROOF** — added on top
5. **DECORATE** — made it ours



posts hold the floor off the ground



me + my brother



we decorated it!

BIRDHOUSE & GLIDER

▪ BIRDHOUSE: FROM BARE WOOD TO FINISHED



▪ BALSA-WOOD GLIDER



Piece by Piece

▪ WHAT IT IS

Building has always been one of my favorite things to do. These photos show me painting a wooden birdhouse and putting together a balsa-wood glider.

▪ MY ROLE

I painted my own design on the birdhouse and assembled the glider piece by piece.

▪ WHY I INCLUDED IT

These early projects are where my love of making things started, and they led to bigger builds like my fort and my go-kart.

MUSIC PRODUCTION

▪ WHAT IT IS

I love playing music. I make and mix music with my MIDI keyboard, my laptop and my DJ controller.

▪ MY ROLE

I've created my own mixes, performed DJ mixing in my school talent show, and played at parties for my neighborhood.

▪ WHY I INCLUDED IT

I like it when people are happy, dancing and enjoying music together.

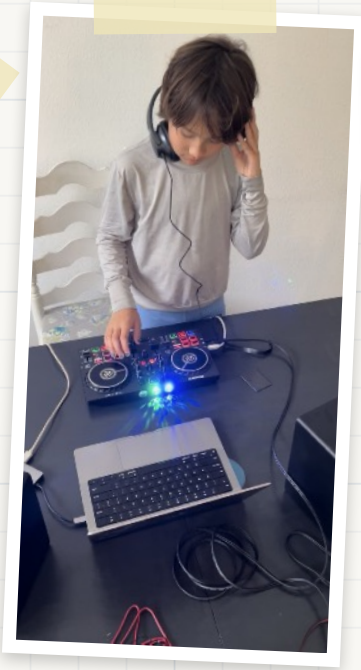
the best part is when everyone starts dancing

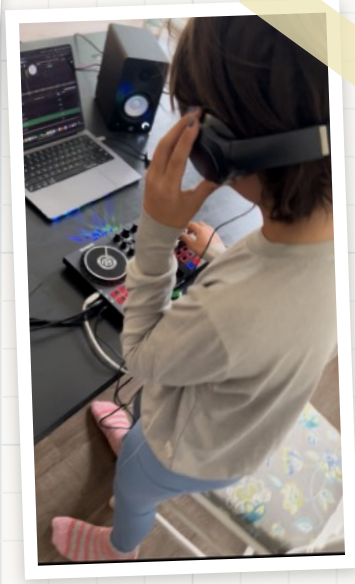
▪ WHERE I'VE PLAYED

TALENT SHOW 2026 | NEIGHBORHOOD PARTIES | HOME STUDIO



my home studio





mixing live







my act was on the program!

19 THEATRE EXPERIENCE • PERFORMANCE

ENCANTO



WHAT IT IS

Encanto was a youth musical theatre production I performed in, with a big cast, bright costumes and a painted set.

MY ROLE

I was part of the cast, singing and acting on stage alongside everyone else to bring the story to life.

WHY I INCLUDED IT

Being on stage showed me how much the set, costumes and lighting shape a story, and that's the side of theatre I want to learn next.









next: designing the set!